Sudhanshu Sakhala

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PROFESSIONAL SUMMARY

AI Engineer with 2+ years of experience architecting and deploying production-grade Agentic AI systems, LLM applications, and enterprise AI platforms. Strong expertise in LangGraph, LangChain, MCP, RAG, LLM orchestration, context engineering, AI memory, and PostgreSQL/pgvector. Experienced in integrating 6+ LLM providers, building 150+ MCP tools, and developing scalable FastAPI-based agent services with streaming, observability, and reliability controls. Proven track record of reducing context usage by ∼93%, token consumption by 65%, and LLM inference costs by 70–80% through intelligent architecture and optimization.

EDUCATION

Dr. Vishwanath Karad MIT World Peace University Aug 2022 – Aug 2024
Master of Computer Application (MCA) Pune, India

- **CGPA:** 8.46 / 10

TECHNICAL SKILLS

Programming Query Languages: Python, SQL, PySpark, Bash, JavaScript, TypeScript

AI LLM Engineering: Agentic AI, LangChain, LangGraph, MCP, ReAct Agents, RAG, LLM Orchestration, Tool Calling, Prompt Engineering, Context Management, Prompt Caching, Token Optimization, OpenAI GPT, Anthropic Claude, Gemini, Google Vertex AI, Qwen

AI Infrastructure Evaluation: Langfuse, LLM Observability, Golden Dataset Evaluation, LLM Routing, Failover, SSE Streaming, Context Isolation, Cost Optimization

Backend Development: FastAPI, REST APIs, AsyncIO, Pydantic, Microservices, Authentication, Authorization, RBAC, Rate Limiting, Session Management, SSE

Data Retrieval: PostgreSQL, pgvector, HNSW, BM25, RRF, Cross-Encoder Reranking, Knowledge Graphs, Redis, MongoDB

Cloud Platforms: Microsoft Azure (ADF, ADLS Gen2, Azure Databricks, Azure Synapse Analytics), Google Cloud Platform (GCP, Vertex AI), AWS, Yeedu

DevOps Tools: Docker, Kubernetes, Azure DevOps, CI/CD, Git, Linux, Portkey, JSON

PROFESSIONAL EXPERIENCE

Modak Analytics (ForgeAI) Nov 2024 – Present
AI Engineer Hybrid

- Architected and shipped ForgeAI, a production AI platform managing 5 MCP servers and 130+ tools integrating 5+ enterprise data sources (GitHub, Jira, Yeedu, Databricks, Reltio) through a unified AI interface.
- Designed a multi-stage LangGraph agent system (Analyst → Data Engineer → Tester) with dynamic tool filtering and context-isolated sub-agents, reducing parent context from 757K to 50K tokens (∼93% reduction).
- Built high-performance RAG and memory systems using PostgreSQL, pgvector, HNSW, BM25, RRF, Cross-Encoder reranking, and knowledge-graph retrieval, achieving sub-300ms query latency.
- Engineered a 5-layer context optimization pipeline with summarization, context editing, tool-result truncation, progressive discovery, and repetition guards, reducing token usage by 65% and enabling 4× larger effective context windows.
- Integrated 6 LLM providers (OpenAI, Claude, Gemini, Qwen, Azure OpenAI, Grok) via Portkey with intelligent routing, failover, and prompt caching, achieving 70–80% LLM cost reduction.

– Built production-grade AI infrastructure with Langfuse observability, Golden Dataset evaluations, SSE streaming, Redis concurrency controls, conversation branching, RBAC, prompt-injection protection, and zero-storage credential forwarding.

Modak Analytics (Yeedu.io)
AI Engineer Hybrid

– Designed and developed Yeedu AssistantX, a production-grade AI assistant for cloud platform operations and developer workflows.
– Built a provider-agnostic LLM orchestration framework supporting OpenAI GPT and Anthropic Claude using LangChain and LangGraph.
– Developed scalable FastAPI microservices with streaming responses, authentication, rate limiting, and Redis-based session management.
– Created and integrated 25+ MCP tools enabling AI agents to interact with clusters, jobs, notebooks, Git repositories, storage systems, and metadata services.
– Implemented prompt caching, token tracking, and conversation summarization, reducing inference costs and improving context efficiency.
– Containerized applications using Docker and enhanced agent reliability through structured error handling, tool recovery, and execution safeguards.

ACHIEVEMENTS

Built and deployed ForgeAI, a production-grade enterprise AI platform integrating 5 MCP servers, 130+ tools, and 5+ enterprise data sources for agent-driven automation.

Engineered multi-agent LangGraph, RAG, memory, and context optimization systems, reducing agent context consumption by $\sim$93%, token usage by 65%, and achieving sub-300ms retrieval latency.

Integrated 6 major LLM providers with intelligent routing, failover, and prompt caching, achieving 70–80% reduction in LLM inference costs.

Developed Yeedu AssistantX with 25+ MCP tools, multi-provider LLM orchestration, FastAPI microservices, streaming, authentication, and Redis-based session management.